and a streaming AI function. It streams the AI response and continuously attempts to parse and validate it. As soon as a valid object is parsed, it's returned to the consumer. This process continues, allowing for multiple objects to be streamed over time.

I do not show an example of how this works since it's identical to the previous example with the `streamText`.

4.3.4 Integrating structured data generation into our web application

Now that we understand these helpers, let's see how we can use them in our project to generate JSON responses instead of text responses. We'll modify our Astra AI agent to provide structured data that we can display as a table. Then, when we ask for a specific product category, the AI agent will respond with a product description.

Here's what we're going to build:

- An updated version of our Astra AI chat application that generates structured JSON data instead of plain-text responses, using the `generateObject` helper function
- A modified AI server action that uses a product schema to request and validate JSON output from the AI model
- A user interface that can display this structured product data in a more organized format using a table

NOTE The source code for the examples in this chapter is available at https://github.com/Generative-AI-Web-Apps/Code/tree/main/ch04/chat-rsc-structured-data.

Project code: chat-rsc-structured-data

The code for this example project is in the ch04/chat-rsc-structured-data folder. You can use that example by running this command in the source folder:

```
$ npm run dev -w ch04/chat-rsc-structured-data
```

You'll also need a Google Gemini API key to access their REST API. The appendix offers instructions on how to set up and create a secret key for Google AI and other providers:

```
// .env
GEMINIGEMINI_API_KEY=<SECRET_KEY>
```

For details on setting up and creating API keys, refer to the appendix.

We will reuse the product schema from the generateObject example described in section 1.3.3. As a reminder, `generateObject` is a function that facilitates the creation of structured data from AI responses. It takes a Zod schema and an AI completion